DIRECTION

Direction Sense: Turns, Distance, Shadows, Clocks

OAVS TGT | Reasoning (Verbal) | Topic 3 of the syllabus list

Syllabus line (Reasoning, 30 Marks): “The Verbal Reasoning topics like Series, Ranks, **Direction**, Arrangement, Coding, Decoding, Analogy and Classification/Odd Pair, Syllogism and Statement Conclusion.”

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). All 9 direction questions found in these papers are solved here.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	Counts, patterns, priority, index of all 9 PYQs
2	Foundation	Compass, left/right table, turn-by-angle method, coordinate method
3	Five Question Types	Walk and distance, turns by angle, shadows, face to face, clocks
4	Shortcuts and Traps	Speed methods and mistakes; scope against Arrangement
5	Practice Set	22 original questions with answer key
6	Quick Revision	One-page cheat sheet

SCOPE: Left/right inside seating arrangements (people sitting in a row or a circle) belongs to **Arrangement**. Letter and number direction in code questions belongs to **Coding**. This PDF covers movement, turning, shadows and clock directions.

SOURCE NOTE: Three Social Studies 2023 questions (Q16, Q19, Q21) printed in the source file have the word “[direction]” in place of a turn. The missing turn is worked out from the official answer in Chapter 3 and marked clearly.

CHAPTER 1

PYQ Analysis

What the five papers tell us about Direction

Paper	Q. numbers	Count	What was asked
English 2019	50, 57	2	Opposite direction of a chess opponent; clock hand direction
English 2022	46, 60	2	Shadow at 9 a.m.; turns of 180°, 225°, 45°
Mathematics 2024	86	1	Two persons' walks; distance between end points
Science 2024	91	1	Five-leg walk; distance and direction back to start
Social Studies 2023	16, 19, 21	3	Walks: direction and distance of the end point
Total		**9**	**Found in all 5 papers; about 2 per paper**

Pattern	PYQs	Priority
Walk with 90° turns: distance and/or direction from start	5	★★★ Must
Turns by angle (180°, 225°, 45°) from a starting direction	1	★★ High
Shadow and sun direction	1	★★ High
Two persons face to face (opposite direction)	1	★★ High
Clock hands as directions	1	★★ High
Expected, not yet asked: Pythagoras end distance, left/right-turn sequences, 8-direction facing after turns	0	★ Expected

PYQ INSIGHT: In **all five** walk questions the final answer lies on a **straight line** with the start or the other end (87 m West, 104 m, 37 m South, 27 km East, 15 km). None needed Pythagoras. The method is to track coordinates and subtract.

PYQ INSIGHT: Every walk question uses only **90° turns**; every angle question uses multiples of 45°. Shadow and clock questions are one-line rules. These are fast marks if the rules are memorised.

Index of the 9 PYQs

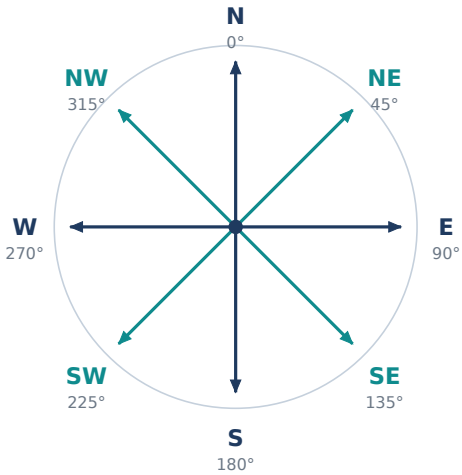
Paper / Q	Question in short	Ans	See
Eng 2019 Q50	Vasanth faces South-East; his chess opponent faces?	North-West	3.4
Eng 2019 Q57	Clock at 3:00; minute hand points North; hour hand?	East	3.5
Eng 2022 Q46	9 a.m., shadow on Amal's left; school direction?	North	3.3
Eng 2022 Q60	Raman faces West; 180° cw, 225° acw, 45° acw	South	3.2
Maths 2024 Q86	Carl and Peter walks; shortest distance between ends	15 km	3.1
Science 2024 Q91	Anant: 43 S, left turns 26, 61, 53, 18; return	27 km east	3.1
SST 2023 Q16	Mitali: 50 E, 75, [turn] 105, 25, 32, 100; shop from home	West, 87 m	3.1
SST 2023 Q19	Shantanu: 25 E, 30, [turn] 44, 30, 35; bank from home	104 m	3.1
SST 2023 Q21	Yukti: 40 N, 28, [turn] 70, 55, 73; reach 27 m east of home	37 m south	3.1

CHAPTER 2

Foundation

Compass, left and right, turns by angle, coordinates

2.1 Compass and bearings



Left / right table (memorise)

Facing	Left	Right	Behind
North	West	East	South
East	North	South	West
South	East	West	North
West	South	North	East

Right turn = 90° clockwise: N → E → S → W → N.
Left turn = 90° anticlockwise: N → W → S → E → N.

Opposite pairs: N ↔ S (0° / 180°), E ↔ W (90° / 270°), NE ↔ SW (45° / 225°), NW ↔ SE (315° / 135°). Add or subtract 180° to get the opposite.

2.2 Turning by an angle (bearing method)

Write the starting direction as a bearing, then add the clockwise turn or subtract the anticlockwise turn. If the result is above 360 subtract 360; if it is below 0 add 360. Convert back to a direction.

Step	Do this	Example
1	N = 0, NE = 45, E = 90, SE = 135, S = 180, SW = 225, W = 270, NW = 315	Facing West = 270
2	Clockwise: **add**. Anticlockwise: **subtract**. Then adjust into 0 to 360.	180° clockwise: 270 + 180 = 450, minus 360 = 90 = East

SHORTCUT: For a sequence of left (L) and right (R) turns, count: **net turns = R − L** (each is 90°). Net 0 = same direction, +1 = right of start, +2 = opposite, +3 (or −1) = left of start.

2.3 Coordinate method for walking problems

Rule	Detail
Axes	East = +x, West = −x, North = +y, South = −y. Start at (0, 0).
Each leg	Facing East: x + d. Facing West: x − d. Facing North: y + d. Facing South: y − d.
After each turn	Update the facing direction using the left / right table before the next leg.
Answer	End point (x, y) gives distance and direction: x only → East/West, y only → North/South; both → diagonal.
Diagonal distance	$\sqrt{(x^2 + y^2)}$. Use triples 3-4-5, 5-12-13, 8-15-17, 7-24-25 and their multiples (6-8-10, 9-12-15, 12-16-20, 15-20-25).
Start from the end	The direction of the start from the end is the **opposite** of the end from the start.

End point	x = 0	x > 0	x < 0
y = 0	At start	East	West
y > 0	North	North-East	North-West
y < 0	South	South-East	South-West

CHAPTER 3

Five Question Types

Rule, shortcut and step-by-step PYQ solutions

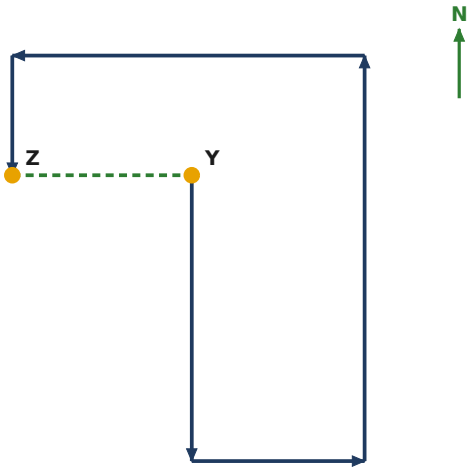
3.1 Type 1: Walk, turns and distance [★★★ five PYQs]

Method: table of Move, Facing, Position. Keep the facing direction in a column so left / right errors show.

Science 2024 Q91: Anant, 43 km south, then four left turns

Anant starts from Y and drives 43 km south, then turns left and drives 26 km, turns left and drives 61 km, turns left and drives 53 km, and takes a final left turn, driving 18 km to Z. How far and in which direction must he go to reach Y?

Move	Facing	Position (x, y)
Start at Y	South	(0, 0)
43 km	South	(0, −43)
Left turn (S → E), 26 km	East	(26, −43)
Left turn (E → N), 61 km	North	(26, 18)
Left turn (N → W), 53 km	West	(−27, 18)
Left turn (W → S), 18 km = Z	South	(−27, 0)

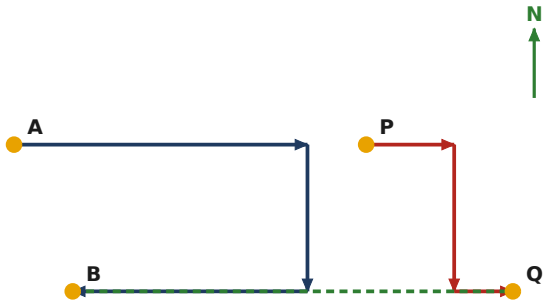


RESULT: Z is at x = −27 and Y is at x = 0, so Z is 27 km **west** of Y. To return he drives **27 km towards the east** (Option 3).

Maths 2024 Q86: Carl and Peter

Carl starts at A and drives 10 km east, turns right and drives 5 km, takes a final right turn and drives 8 km to B. Peter starts at P, which is 12 km east of A, drives 3 km east, turns right and drives 5 km, takes a final left turn and drives 2 km to Q. What is the shortest distance between B and Q?

Move	Facing	Position (x, y)
Carl: 10 km east from A (0, 0)	East	(10, 0)
Right turn (E → S), 5 km	South	(10, −5)
Right turn (S → W), 8 km = B	West	(2, −5)
Peter: 3 km east from P (12, 0)	East	(15, 0)
Right turn (E → S), 5 km	South	(15, −5)
Left turn (S → E), 2 km = Q	East	(17, −5)



RESULT: B (2, −5) and Q (17, −5) have the **same y**, so the distance is 17 − 2 = **15 km** (Option A). No Pythagoras needed.

SST 2023 Q16, Q19, Q21 (the printed turn is missing)

SOURCE NOTE: The source text of these three questions says “turns [direction]”. The word was lost. Only one choice reproduces the official answer, so that choice is used below. Learn the **method**; the missing turn does not change it.

Q16. Mitali walks 50 m east, turns right and walks 75 m, turns [**right**] and walks 105 m, turns left and walks 25 m, turns right and walks 32 m, turns right and walks 100 m to a shop. Direction and distance of the shop from home?

Move	Facing	Position (x, y)
50 m	East	(50, 0)
Right (E → S), 75 m	South	(50, −75)
[Right] (S → W), 105 m	West	(−55, −75)
Left (W → S), 25 m	South	(−55, −100)
Right (S → W), 32 m	West	(−87, −100)
Right (W → N), 100 m = shop	North	(−87, 0)

RESULT: Shop at (−87, 0): **West, 87 m** (Option 4).

Q19. Shantanu walks 25 m east, turns right and walks 30 m, turns [**left**] and walks 44 m, turns left and walks 30 m, finally turns right and walks 35 m to a bank. How far is the bank from home?

Move	Facing	Position (x, y)
25 m	East	(25, 0)
Right (E → S), 30 m	South	(25, −30)
[Left] (S → E), 44 m	East	(69, −30)
Left (E → N), 30 m	North	(69, 0)
Right (N → E), 35 m = bank	East	(104, 0)

RESULT: Bank at (104, 0): **104 m** from home (Option 4).

Q21. Yukti walks 40 m north, turns left and walks 28 m, turns [**right**] and walks 70 m, turns right and walks 55 m, turns right and walks 73 m. In which direction and how far must she walk now to reach a point 27 m east of her home?

Move	Facing	Position (x, y)
40 m	North	(0, 40)
Left (N → W), 28 m	West	(−28, 40)
[Right] (W → N), 70 m	North	(−28, 110)
Right (N → E), 55 m	East	(27, 110)
Right (E → S), 73 m	South	(27, 37)

RESULT: Target = (27, 0). She is at (27, 37), so she must walk **37 m south** (Option 3).

TRAP: A wrong left / right turn changes every later coordinate. Write the **facing direction** next to each leg before computing.

3.2 Type 2: Turning by an angle [★★]

Method: bearing arithmetic (clockwise +, anticlockwise −).

PYQ Eng 2022 Q60 Three turns by angle

Q. Raman is facing west. He turns 180° clockwise, then 225° anticlockwise, then 45° anticlockwise. Which direction is he facing now?

Step 1. West = 270°. Turn 180° clockwise: 270 + 180 = 450, minus 360 = 90° (East).
Step 2. 225° anticlockwise: 90 − 225 = −135, plus 360 = 225° (South-West).
Step 3. 45° anticlockwise: 225 − 45 = 180° (South).
Answer: South (Option B)

SHORTCUT: Add the net turn first: 180 clockwise − 225 − 45 = −90, which is **90° anticlockwise** overall. From West, 90° anticlockwise is South. One calculation instead of three.

3.3 Type 3: Shadows and the sun [★★]

Time	Sun is in	Shadow falls to	Person facing the sun faces
Morning (sunrise)	East	West	East
Evening (sunset)	West	East	West

Where the shadow is relative to the person	Morning: facing	Evening: facing
Shadow on the left	North	South
Shadow on the right	South	North
Shadow behind	East	West
Shadow in front	West	East

PYQ Eng 2022 Q46 Shadow at 9 a.m.

Q. Amal is going to school at 9 a.m. He observes his shadow to his left. In which direction is his school from his house?

Step 1. At 9 a.m. the sun is in the east, so his shadow falls to the west.
Step 2. The shadow (west) is on his left, so his left is west. Facing North: left = West.
Step 3. He faces north, and he is walking to school, so the school is in the north.
Answer: North (Option C)

SHORTCUT: Morning and shadow on left: facing North. Use the table and answer in 5 seconds.

3.4 Type 4: Face to face [★★]

Rule: two persons facing each other face **opposite** directions. A's right hand is B's left hand.

PYQ Eng 2019 Q50 Chess opponents

Q. Vasanth and Ramesh are playing chess sitting in standard seating positions. If Vasanth faces South-East, which direction is his opponent Ramesh facing?

Step 1. Players sit facing each other, so Ramesh faces the opposite direction.
Step 2. Opposite of South-East is North-West (135° + 180° = 315°).
Answer: North-West (Option C)

3.5 Type 5: Clock directions [★★]

Rule: the direction the minute hand points is '12'. The hour hand at an exact hour is turned **hour × 30°** clockwise from the minute hand.

Hour	3:00	6:00	9:00
Hour hand relative to the minute hand	90° clockwise	180° (opposite)	270° clockwise = 90° anticlockwise

PYQ Eng 2019 Q57 Clock at 3:00

Q. The clock shows 3:00. If the minute hand points towards North, in which direction does the hour hand point?

Step 1. At 3:00 the minute hand is on 12 (North) and the hour hand is on 3.
Step 2. From 12 to 3 is 90° clockwise, which is North → East.
Answer: East (Option A)

SHORTCUT: If the minute hand points North: at 3:00 the hour hand points East, at 6:00 South, at 9:00 West. If the minute hand points *another* direction, rotate the answers by the same angle.

CHAPTER 4

Shortcuts and Traps

Marks saved and marks lost

- **Always write the facing direction** next to each leg of a walk.
- **Net turns:** count R minus L, then use the table (0 = same, 1 = right, 2 = opposite, 3 = left).
- **Angle questions:** add all clockwise and subtract all anticlockwise first, then apply it once.
- **Shadow:** morning shadow = West, evening shadow = East. Memorise the left / right table for the shadow.
- **Opposite direction** = add 180°: NE ↔ SW, NW ↔ SE.
- **Walk answers are often on one line** (all five PYQs): subtract coordinates instead of using Pythagoras.
- **Pythagoras triples:** 3-4-5, 5-12-13, 8-15-17, 7-24-25 and their multiples.

Trap	How to avoid it
Turning ‘left’ when facing South	Facing South, left = East (not West). Use the table; do not use the picture in your head.
Confusing ‘direction of A from B’ with ‘B from A’	They are opposite. Re-read which person is the reference.
Using distance walked instead of displacement	‘How far from the start’ means the straight-line distance, not the total walked.
Clock hand at an exact hour only	At times like 3:30 the hour hand is between numbers; PYQs used only exact hours.
Missing word in the question	If a printed turn is blank, ask for the answer option that works; here the method still scores.
Angle sign error	Clockwise adds, anticlockwise subtracts. West minus 90° is South, not North.

Scope check

PYQ	Why it is not covered here
Maths 2024 Q73, Q74; Science 2024 Q88, Q93; Eng 2022 Q34, Q37, Q42–Q49, Q55; SST 2023 Q6, Q17, Q27	People facing a direction inside a seating arrangement: Arrangement
Direction facing in Science 2024 Q88 (north) and SST 2023 Q6	Used only to define left and right inside a row: Arrangement

CHAPTER 5

Practice Set

22 original questions; answers on the next page

Q	Question
1	Rahul faces North. He turns right, then right, then left. Which direction does he face?
2	Priya faces South. She turns left, left, right, right and left. Which direction does she face?
3	A man faces East and turns 135° anticlockwise. Which direction does he face?
4	A girl faces North-West. She turns 90° clockwise and then 135° anticlockwise. Which direction does she face?
5	A boy faces South and turns 270° clockwise. Which direction does he face?
6	A woman faces West. She turns 45° clockwise and then 180°. Which direction does she face?
7	Ravi walks 12 m north, turns right and walks 5 m. How far and in which direction is he from the starting point?
8	Seema walks 8 km south, turns left and walks 6 km, turns left and walks 8 km. How far and in which direction is she from the start?
9	Deepa walks 15 m west, turns right and walks 12 m, then turns right and walks 6 m. How far and in which direction is she from the start?
10	A walks 6 km east and then 8 km north. B walks 6 km west and then 8 km south. What is the distance between them?
11	Meera walks 20 m south, turns left and walks 30 m, turns left and walks 35 m, then turns left and walks 30 m. How far and in which direction is she from the start?
12	In the morning, Rohit’s shadow falls on his right. Which direction is he facing?
13	In the evening, Sima’s shadow falls behind her. Which direction is she facing?
14	At sunrise, Arun stands facing the sun. In which direction does his shadow fall?
15	In the evening, Mohan’s shadow falls on his left. Which direction is he facing?
16	A faces North-East and B sits opposite to A, facing A. Which direction does B face?
17	A faces West. B is facing A. In which direction is B’s right hand?
18	At 6:00 the minute hand points East. In which direction does the hour hand point?
19	At 9:00 the minute hand points South. In which direction does the hour hand point?
20	At 3:00 the minute hand points West. In which direction does the hour hand point?
21	A man faces North-East. He turns 90° anticlockwise and then 225° clockwise. Which direction does he face?
22	A man faces South-East. He turns 135° clockwise and then another 135° clockwise. Which direction does he face?

Answer key with logic

Q	Answer	Logic
1	East	Net turns = 2 right − 1 left = +1: right of North = East
2	East	Right 2 − Left 3 = −1: left of South = East
3	North-West	East = 90; 90 − 135 = −45 → 315
4	West	315 + 90 = 405 → 45; 45 − 135 = −90 → 270
5	East	180 + 270 = 450 → 90
6	South-East	270 + 45 = 315; 315 + 180 = 495 → 135
7	13 m, North-East	Position (5, 12); √(25 + 144) = 13; both x and y positive
8	6 km, East	(0, −8) → (6, −8) → (6, 0)
9	15 m, North-West	(−15, 0) → (−15, 12) → (−9, 12); √(81 + 144) = 15
10	20 km	A (6, 8), B (−6, −8); √(12² + 16²) = 20
11	15 m, North	(0, −20) → (30, −20) → (30, 15) → (0, 15)
12	South	Morning shadow West on the right: facing South
13	West	Evening shadow East; shadow behind means facing West
14	West	Sun in the East at sunrise; shadow West
15	South	Evening shadow East is on his left: facing South
16	South-West	Opposite of NE
17	South	B faces East; B’s right is South
18	West	Hour hand is opposite the minute hand at 6:00
19	East	9:00 is 90° anticlockwise from 12; from South that is East
20	North	3:00 is 90° clockwise from 12; from West that is North
21	South	45 − 90 = −45 → 315; 315 + 225 = 540 → 180
22	North-East	135 + 270 = 405 → 45

CHAPTER 6

Quick Revision

One page for the last day

Question asks	Use	PYQ check
Walk, final position	Coordinate table with facing direction	Science 2024 Q91: Z at (−27, 0); 27 km east to return
Distance between two ends	Subtract coordinates; Pythagoras if both differ	Maths 2024 Q86: 17 − 2 = 15 km
Turn by angle	Bearing: clockwise +, anticlockwise −	Eng 2022 Q60: 270 → 90 → 225 → 180 = South
Sequence of left / right turns	Net turns = R − L	0 same, 1 right, 2 opposite, 3 left
Shadow	Morning shadow West; evening shadow East	Eng 2022 Q46: shadow on left → North
Face to face	Opposite direction	Eng 2019 Q50: SE ↔ NW
Clock	Hour hand = hour × 30° clockwise from the minute hand	Eng 2019 Q57: 3:00, N → East

Facing	Left	Right	Behind
North	West	East	South
East	North	South	West
South	East	West	North
West	South	North	East

ROUTINE: Exam-day routine: write the facing direction beside every leg, update (x, y), read the answer from the final coordinates. Angles: bearings. Shadows: sun East in the morning. 30 seconds per question.